

# Add, Subtract and Multiply Complex Numbers

## Key Points

To add complex numbers, add the real parts and then add the imaginary parts

$$(a + bi) + (c + di) = a + c + (b + d)i$$

To subtract complex numbers, subtract the real parts and then subtract the imaginary parts

$$(a + bi) - (c + di) = a - c + (b - d)i$$

To multiply complex numbers, use FOIL(or quadratic multiplication)

$$(a + bi)(c + di) = ac - bd + (ad + bc)i$$

Remember that  $i^2 = -1$

## Basic Skills

**Q1.** For  $z = 3 + 2i$ ,  $w = 2 + i$  and  $v = 2 - 8i$  calculate the following

- a)  $z + w$
- b)  $z - w$
- c)  $z \times w$
- d)  $w \times v$
- e)  $w - v$
- f)  $5w$
- g)  $z - 2w$

## Competency Skills

**Q2.** For  $z = -3 + 5i$ ,  $w = -(5 + i)$  and  $v = -2 + i$  calculate the following

- a)  $2z + w$
- b)  $z - 2w$
- c)  $z w + v$
- d)  $w^2 + z$
- e)  $v^2 - w$
- f)  $(2z)^2$
- g)  $3z^2 + 2z - 4$

**Q3.** For  $z = 3i - 1$  find  $z^2 - 2z$

**Q4.** What is  $\text{Re}(z^3)$  where  $z = 5 + 2i$

**Q5.** A quadratic equation has  $z^2 + 25 = (z + ai)(z + bi)$ . What are  $a$  and  $b$ ?

## Advanced Skills

**Q6.** State the value of  $\text{Im}((5 + i)^2)$  and determine if  $\text{Im}(z^2) = \text{Im}(z)^2$

**Q7.** A quadratic equation has  $z^2 - 3iz + 25 = (z + ai)(z + bi)$ . What are  $a$  and  $b$ ?

**Q8.** For  $z = 3 - 5i$ ,  $w = 2i - 5$  calculate the following values

- a)  $z^2 - 2iw$
- b)  $(i + w)^2 + z$

**Q9.** If  $z = a + bi$  and  $z^* = a - bi$  for some real constants  $a, b$  determine in terms of  $a, b$  the product of  $z$  and  $z^*$  (i.e  $zz^*$ )

## Extension

**Q10.** Find all solutions to the equation  $z^4 = 4$